

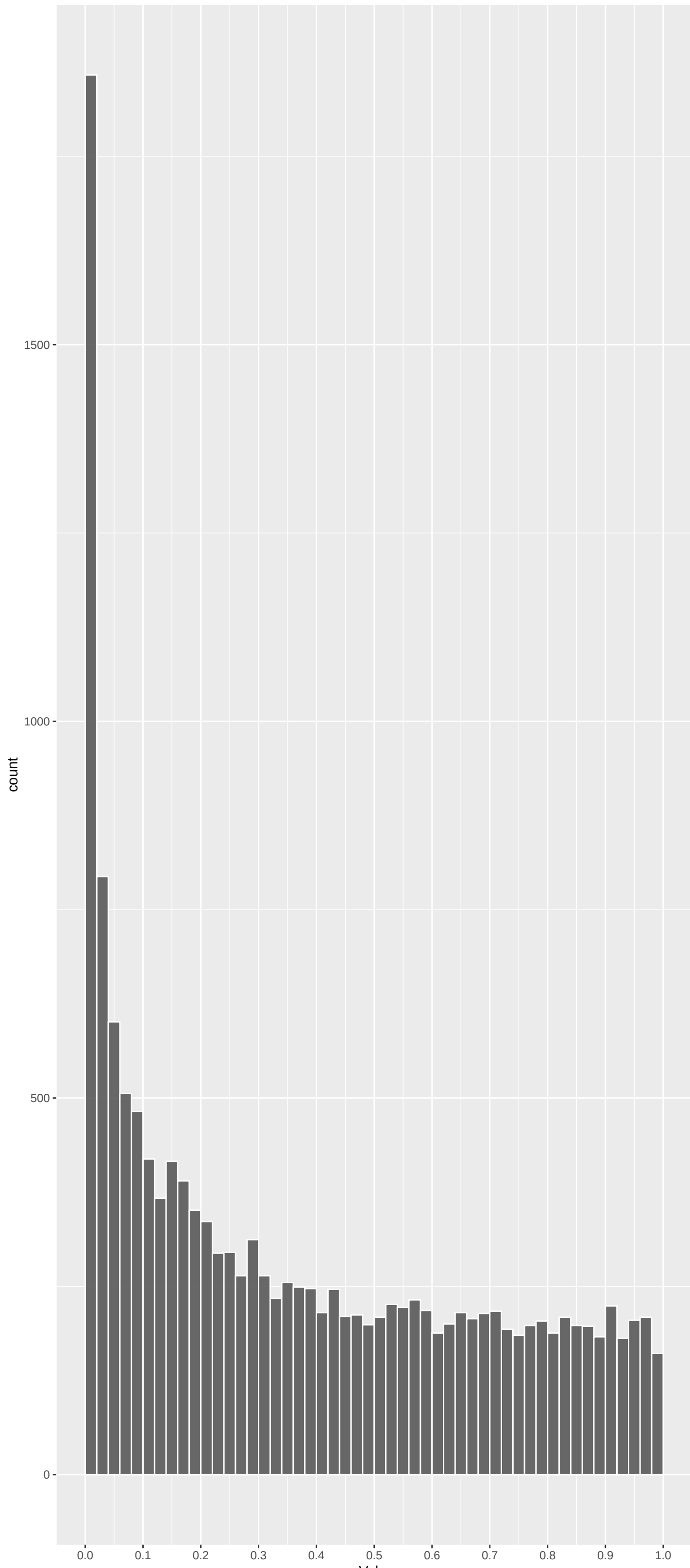
Top genes by P-value non-permuted

Gene	Rho	N	P	p.adj	qValueNoperm
PKD1L1	0.07943546	285	2.037e-06	4.567e-03	9.998e-01
ANOS	0.07146647	339	8.885e-07	4.567e-03	9.998e-01
SULT1A2	0.07235461	303	2.946e-06	4.567e-03	9.998e-01
R3HDM4	0.09033655	268	9.052e-07	4.567e-03	9.998e-01
H3-3A	0.28229343	85	1.118e-06	4.567e-03	9.998e-01
DYNLL1	0.24836512	91	2.269e-06	4.567e-03	9.998e-01
PROM1	0.06324263	370	1.360e-06	4.567e-03	9.998e-01
TRA2B	0.22598385	98	2.842e-06	4.567e-03	9.998e-01
TANGO2	0.07956759	275	3.024e-06	4.567e-03	9.998e-01
RPL23	0.24246721	92	2.636e-06	4.567e-03	9.998e-01
PPP4C	0.21945566	96	4.971e-06	6.257e-03	9.998e-01
IQGAP3	0.05817292	360	4.879e-06	6.257e-03	9.998e-01
TUBB8	0.24559162	85	5.573e-06	6.474e-03	9.998e-01
SPINT4	0.07351568	278	6.403e-06	6.766e-03	9.998e-01
TUBB8B	0.23550962	87	6.782e-06	6.766e-03	9.998e-01
DNMT3B	0.06223098	319	8.646e-06	6.766e-03	9.998e-01
SLC22A16	0.06726364	293	9.344e-06	6.766e-03	9.998e-01
GLRA4	0.07713823	261	7.521e-06	6.766e-03	9.998e-01
RPS14	0.21522807	93	8.594e-06	6.766e-03	9.998e-01
TRPV1	0.06244733	317	8.903e-06	6.766e-03	9.998e-01
SULT1A3	0.06133837	321	9.407e-06	6.766e-03	9.998e-01
HIRA	0.06767517	280	1.391e-05	7.084e-03	9.998e-01
CD80	0.07266595	262	1.331e-05	7.084e-03	9.998e-01
UBE2B	0.19831177	98	1.155e-05	7.084e-03	9.998e-01
DACH2	0.07386174	258	1.319e-05	7.084e-03	9.998e-01

Top genes by Q-Value non-permuted

Gene	Rho	N	P	p.adj	qValueNoperm
M6PR	7.001042e-04	282	6.574e-01	8.487e-01	9.998e-01
FKBP4	6.598911e-04	320	6.464e-01	8.435e-01	9.998e-01
CYP26B1	1.341676e-03	230	5.794e-01	8.029e-01	9.998e-01
NDUFAF7	2.486996e-04	329	7.752e-01	9.081e-01	9.998e-01
FUCA2	4.385138e-03	350	2.187e-01	5.084e-01	9.998e-01
DBNDD1	1.215051e-03	277	5.625e-01	7.939e-01	9.998e-01
HS3ST1	9.734053e-03	270	1.056e-01	3.653e-01	9.998e-01
SEMA3F	6.084565e-03	296	1.803e-01	4.664e-01	9.998e-01
CFTR	2.346800e-05	357	9.272e-01	9.717e-01	9.998e-01
CYP51A1	5.141350e-02	298	9.319e-05	1.857e-02	9.998e-01
USP28	1.783995e-03	343	4.347e-01	7.036e-01	9.998e-01
SLC7A2	9.590228e-03	320	8.028e-02	3.217e-01	9.998e-01
HSPB6	3.115323e-03	232	3.963e-01	6.735e-01	9.998e-01
PKD4	1.031645e-02	284	8.751e-02	3.348e-01	9.998e-01
USH1C	5.343154e-03	348	1.733e-01	4.614e-01	9.998e-01
BAIAP2L1	7.607801e-03	316	1.216e-01	3.921e-01	9.998e-01
TMEM132A	2.232753e-02	349	5.312e-03	8.783e-02	9.998e-01
DVL2	6.676109e-04	280	6.660e-01	8.534e-01	9.998e-01
RPAP3	3.270712e-03	337	2.945e-01	5.857e-01	9.998e-01
HOXA11	2.314794e-04	207	8.271e-01	9.313e-01	9.998e-01
TRAPPC6A	1.388635e-03	262	5.472e-01	7.841e-01	9.998e-01
SPATA20	2.083751e-04	347	7.883e-01	9.143e-01	9.998e-01
CD79B	1.546554e-03	348	4.638e-01	7.265e-01	9.998e-01
RHBDD2	3.842415e-07	271	9.919e-01	9.967e-01	9.998e-01
RPUSD1	6.915815e-03	295	1.539e-01	4.363e-01	9.998e-01

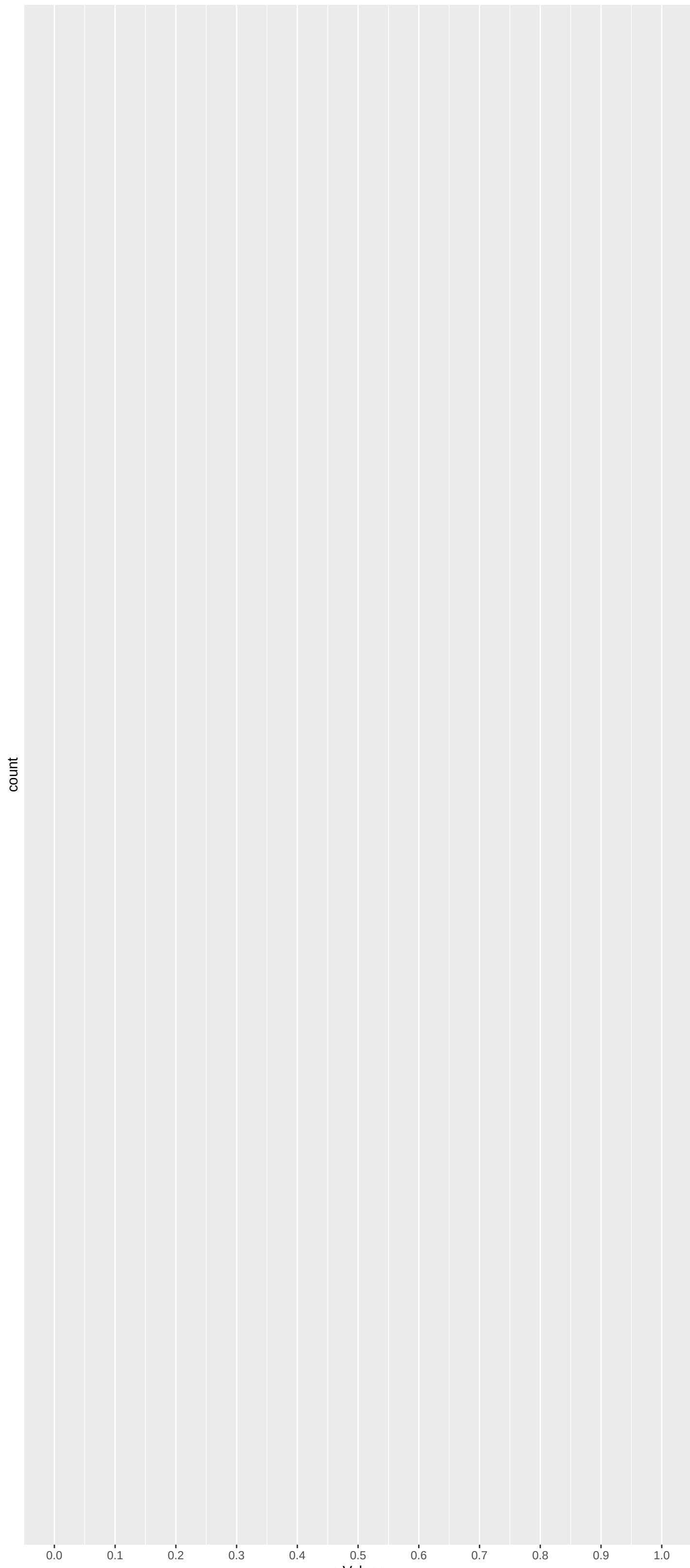
Positive Rho Non-permuted



Top Positive genes by P-value non-permuted

Gene	Rho	N	P	p.adj	qValueNoperm
PKD1L1	0.07943546	285	2.037e-06	4.567e-03	9.998e-01
ANOS	0.07146647	339	8.885e-07	4.567e-03	9.998e-01
SULT1A2	0.07235461	303	2.946e-06	4.567e-03	9.998e-01
R3HDM4	0.09033655	268	9.052e-07	4.567e-03	9.998e-01
H3-3A	0.28229343	85	1.118e-06	4.567e-03	9.998e-01
DYNLL1	0.24836512	91	2.269e-06	4.567e-03	9.998e-01
PROM1	0.06324263	370	1.360e-06	4.567e-03	9.998e-01
TRA2B	0.22598385	98	2.842e-06	4.567e-03	9.998e-01
TANGO2	0.07956759	275	3.024e-06	4.567e-03	9.998e-01
RPL23	0.24246721	92	2.636e-06	4.567e-03	9.998e-01

Negative Rho Non-permuted



Top Negative genes by P-value non-permuted

Gene	Rho	N	P	p.adj	qValueNoperm
NA	NA	NA	NA	NA	NA
NA.1	NA	NA	NA	NA	NA
NA.2	NA	NA	NA	NA	NA
NA.3	NA	NA	NA	NA	NA
NA.4	NA	NA	NA	NA	NA
NA.5	NA	NA	NA	NA	NA
NA.6	NA	NA	NA	NA	NA
NA.7	NA	NA	NA	NA	NA
NA.8	NA	NA	NA	NA	NA
NA.9	NA	NA	NA	NA	NA

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Positive Regulation Of Cytokine Product	0.08873829	262	8.673e-07	4.672e-03	CD80:27 SYK:48 CD86:55 IL7:138 TNF:189 NLRP9:217
Cytoplasmic Translation (GO:0002181)	0.22112506	39	1.793e-06	4.828e-03	RPS14:17 ETF1:59 RPS15A:79 RACK1:106 RPS27:207 RPL19:252
Gene Expression (GO:0010467)	0.08599387	205	2.363e-05	2.546e-02	RPS14:17 RPS15A:79 CASP3:112 MAGOH:176 RPS27:207 RNASEL:227
Peptide Biosynthetic Process (GO:0043043)	0.12423791	99	2.009e-05	2.546e-02	RPS14:17 GTTLC3:68 RPS15A:79 GTTLC1:194 RPS27:207 DMD:244
Proton Motive Force–Driven Mitochondrial	0.18036291	47	1.915e-05	2.546e-02	NDUFS5:43 NDUFAl1:242 NDUFAl2:363 NDUFB5:753 NDUFS3:808 NDUFB3:851
Oxidative Phosphorylation (GO:0006119)	0.16271677	54	3.590e-05	3.224e-02	NDUFS5:43 NDUFAl1:242 NDUFAl2:363 NDUFB5:753 NDUFS3:808 NDUFB3:851
Nervous System Development (GO:0007399)	-0.06252316	363	4.866e-05	3.744e-02	FZD9:41 RBF0X3:73 RITA1:117 LRCH4:142 TMEM223:145 EPHB1:276
Aerobic Electron Transport Chain (GO:001	0.15915949	52	7.784e-05	3.934e-02	NDUFS5:43 COXA5:455 COXB5:535 NDUFB5:753 NDUFS3:808 NDUFB3:851
Mitochondrial ATP Synthesis Coupled Elec	0.16085264	52	6.097e-05	3.934e-02	NDUFS5:43 NDUFAl1:242 COXA5:455 COXB5:535 NDUFB5:753 NDUFS3:808
Organic Hydroxy Compound Transport (GO:0	0.19382903	35	7.302e-05	3.934e-02	SLC44A2:425 SLC44A1:477 ATPB81:508 ABCCL1:538 SLC01A2:678 SLC01B1:684
Leukotriene D4 Biosynthetic Process (GO:	0.45342092	6	1.199e-04	5.754e-02	GTTLC3:68 GTTLC1:194 GGT5:320 GGT6:744 GTTLC2:1109 GGT17:1695
Negative Regulation Of Gene Expression (	0.06771009	273	1.282e-04	5.754e-02	ENG:22 ATP2B4:54 WAC1:109 TASOR:135 SLFN14:187 TNF:189
Positive Regulation Of Macromolecule Met	0.06469852	296	1.408e-04	5.833e-02	DNMT3B:18 FGF21:104 CLU:116 CALCR:144 TNF:189 ACTC1:210
Leukotriene D4 Metabolic Process (GO:190	0.37817833	8	2.123e-04	7.925e-02	GTTLC3:68 GTTLC1:194 GGT5:320 GGT6:744 GTTLC2:1109 GGT1:1121
Regulation Of Synaptic Transmission, Glu	-0.15586869	47	2.207e-04	7.925e-02	HCN1:160 NLGN2:287 CCR2:596 NRXN1:607 NLGN3:733 GRIN2D:943
Mitochondrial Electron Transport, NADH T	0.19041138	31	2.449e-04	8.246e-02	NDUFS5:43 NDUFB5:753 NDUFS3:808 NDUFB3:851 NDUFAl3:991 NDUFAl5:1076
Regulation Of Chemokine Production (GO:0	0.17095358	38	2.679e-04	8.490e-02	IL7:138 TNF:189 IFNG:226 TLR7:279 IL3:297 EIF2AK2:305
RNA Processing (GO:0006396)	0.09501374	122	2.988e-04	8.665e-02	SNRPD1:31 CDK12:159 SRSF1:163 MAGOH:176 RNASEL:227 U2AF1:231
Positive Regulation Of Tumor Necrosis Fa	0.12866120	66	3.056e-04	8.665e-02	SYK:48 CD86:55 CLU:116 IFNG:226 IL7:138 TLR9:443
Proton Motive Force–Driven ATP Synthesis	0.14560129	51	3.257e-04	8.772e-02	NDUFS5:43 NDUFAl1:242 NDUFAl2:363 NDUFB5:753 NDUFS3:808 NDUFB3:851
Defense Response To Symbiont (GO:0140546	0.10667803	93	3.861e-04	9.903e-02	SERINC3:169 NLRP9:217 RNASEL:227 STAT2:235 TLR7:279 EIF2AK2:305
Macrophage Differentiation (GO:0030225)	0.22569397	20	4.771e-04	1.117e-01	IFNG:226 CDC42:228:5 IL33:297 IL31RA:402 LIF:670 CASP10:994
Positive Regulation Of Chemokine Product	0.15597622	42	4.736e-04	1.117e-01	SYK:48 IL7:138 TNF:189 IFNG:226 TLR7:279 IL33:297
Bile Acid And Bile Salt Transport (GO:00	0.25092320	16	5.120e-04	1.120e-01	ATP8B1:508 ABCCL1:538 SLC01A2:678 SLC01B1:684 SLC5A1:865 ABCD3:1015
Positive Regulation Of Nervous System De	-0.159306545	43	5.196e-04	1.120e-01	EPHB1:276 NLGN2:287 SERPINF1:304 ZNF335:457 NRXN1:607 SLITRK1:727
Positive Regulation Of NF-kappaB Transcr	0.08902680	126	5.754e-04	1.122e-01	CLU:116 TNF:189 AKR1B1:271 CARD14:275 EIF2AK2:305 IL18RAP:339
Aerobic Respiration (GO:0009060)	0.35757575	53	6.360e-04	1.224e-01	NDUFS5:43 NDUFAl1:242 NDUFAl2:363 NDUFB5:753 NDUFS3:808 NDUFB3:851
Myeloid Leukocyte Differentiation (GO:00	0.24897566	59	6.193e-04	1.224e-01	IFNG:226 CDC42:228:5 IL33:297 IL4:322 IL31RA:402 GLO1:445
Dendritic Cell Differentiation (GO:00970	0.12582492	16	6.646e-04	1.235e-01	CLU:116 FLT3:157 IL4:322 LYN:1411 CCR7:2895 BAF3:3011:5
Histone Ubiquitination (GO:0016574)	0.34559514	8	7.122e-04	1.279e-01	UBE2B:23 UBR2:90 WAC1:187 RNFA0:2232 UBE2A:2966 UBE2N:148
Sphingolipid Biosynthetic Process (GO:00	0.11787975	68	7.868e-04	1.367e-01	ST8SIA5:96 ELOVL7:151 ST8SIA1:196 HACD:2420 SPTLC3:269 PLP3:280
Steroid Hormone Biosynthetic Process (GO	0.24411449	16	8.421e-04	1.418e-01	HSD17B2:62 CYP17A1:145 HSD17B11:270 CYP19A1:506 HSD17B3:600 HSD3B2:815
Negative Regulation Of Rho Protein Signa	0.25641478	14	8.957e-04	1.419e-01	DYNLL1:6 RASPI1:1136 KANK1:1143 MYOC:1732 RIPOR2:1764 FLCN:2034
Positive Regulation Of Leukocyte Cell-Ce	0.17558557	30	8.776e-04	1.419e-01	TNF:189 PRKCQ:466 ETS1:1131 CDC8B8:1315 RHOA:1367 FCHO1:1868
Positive Regulation Of Tyrosine Phosphor	0.13925232	47	9.659e-04	1.487e-01	FLT3:157 TNF:189 IFNG:226 PARP1:431 IL4:322 IL31RA:402
Positive Regulation Of Tumor Necrosis Fa	0.11909814	62	1.197e-03	1.791e-01	SYK:48 CLU:116 IFNG:226 IL33:297 TLR7:279 TNF:189 ORM2:626
Positive Regulation Of Interleukin-6 Pro	0.11807126	62	1.319e-03	1.920e-01	SYK:48 TNF:189 IFNG:226 TLR7:279 IL33:297 TLR8:334
Negative Regulation Of Type I Interferon	0.17377900	28	1.465e-03	2.077e-01	KLHL22:359 ITCH:569 REL:703 ATG5:1438 RBL1:606 GPATCH3:1663
Calcium–Mediated Signaling Using Intrac	0.21855018	17	1.814e-03	2.334e-01	CLU:116 GPR143:170 VCAM1:370 DMTN:1583 CCL20:2641 PTGFR:3062
Leukotriene Biosynthetic Process (GO:001	0.22576204	16	1.772e-03	2.334e-01	GTTLC3:68 GTTLC1:194 GGT5:320 GGT6:744 MGST3:823 GTTLC2:1109

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_INFLUENZA_INFECTION	0.16934998	82	1.175e-07	2.351e-04	RPL23:7 RPS14:17 RPS15A:79 RPS27:207 POLR23:246 RPL19:252
REACTOME_INFECTIONS_DISEASE	0.06166290	671	6.459e-08	2.351e-04	ANOS1:4 DYNLL1:6 RPL23:7 TUBB8:13 TUBB8:15 RPS14:17
REACTOME_SARS_COV_1_HOST_INTERACTIONS	0.20580913	57	7.786e-08	2.351e-04	RPS14:17 RPS15A:79 UBE2I:94 RPS27:207 TLR7:279 RPS7:374
FLECHNER_BIOPSY_KIDNEY_TRANSPLANT_OK_VS_	0.07413660	432	1.449e-07	2.351e-04	DYNLL1:6 SULT1A2:9 TSPAN8:28 ETF1:59 SCFD1:92 RACK1:109
REACTOME_CELLULAR_RESPONSES_TO_STIMULI	0.06250254	589	2.626e-07	3.407e-04	H3-3A:3 DYNLL1:6 RPL23:7 TUBB8:13 TUBB8:15 RPS14:17
MARSON_BOUND_BY_FOXP3_STIMULATED	0.04856589	854	1.826e-06	1.626e-03	UBE2D1:37 TMEM50A:52 ETF1:59 FBXW12:67 H4C15:103 MYL4:120
REACTOME_SARS_COV_1_MODULATES_HOST_TRANS	0.35623684	15	1.781e-06	1.626e-03	RPS14:17 RPS15A:79 RPS27:207 RPS7:374 RPS26:481 RPS15:597
PUJANA_BRCA1_PCC_NETWORK	0.04090958	1283	2.005e-06	1.626e-03	RPL23:7 TRA2B:8 HIRA:29 SNRPD1:31 MKV:33 HK2:42
WP_NONALCOHOLIC_FATTY_LIVER_DISEASE	0.12297172	123	2.545e-06	1.834e-03	NDUFS5:43 CASP3:112 BID:179 TNF:189 CDC42:228:5 NDUFAl1:242
REACTOME_EUKARYOTIC_TRANSLATION_ELONGATI	0.21074641	41	3.043e-06	1.855e-03	RPL23:7 RPS14:17 RPS15A:79 EEF1B2:121 RPS27:207 RPL19:252
REACTOME_SARS_COV_1_INFECTION	0.13997249	93	3.146e-06	1.855e-03	RPS14:17 RPS15A:79 UBE2I:94 RPS27:207 RPS7:374 RPS27:207 TLR7:279
MULLIGHAN_NPM1_SIGNATURE_3_UP	0.08137843	275	3.641e-06	1.968e-03	UBE2K:58 UBR2:90 SERINC3:169 ST3GAL2:177 APC2:286 SLC46A3:365
WP_NETWORK_MAP_OF_SARSCOV2_SIGNALING_PAT	0.10270600	165	5.517e-06	2.722e-03	FGG:30 CASP3:112 APOD1:114 CXCL13:126 TNFSF10:127 IL7:138
SCHLOSSER_SERUM_RESPONSE_DN	0.05654773	556	5.875e-06	2.722e-03	TRA2B:8 UBR2:80 UBE2G1:90 PPP3CC:107 RBBP4:115 EEF1B2:121
KEGG_HUNTINGTONS_DISEASE	0.11038698	128	1.653e-05	7.147e-03	NDUFS5:43 CASP3:112 TFAM:185 POLR2J2:46 SLC25A4:308 COXA5:455
REACTOME_MRNA_SPLICING_MINOR_PATHWAY	0.24516817	25	2.209e-05	8.495e-03	SNRPD1:31 SF3B5:122 SF3B6:128 RPS17:163 POLR2J2:46 SNRPD3:534
HSIAO_HOUSEKEEPING_GENES	0.07646127	261	2.226e-05	8.495e-03	H3-3A:3 DYNLL1:6 RPL23:7 RPS14:17 RPS15A:79 JUND:83
KEGG_RIBOSOME	0.20052473	37	2.444e-05	8.809e-03	RPL23:7 RPS15A:79 RPS27:207 RPL19:252 RPS7:374 MRPL13:446
YAO_TEMPORAL_RESPONSE_TO_PROGESTERONE_CL	0.10861709	125	2.805e-05	9.579e-03	GATAD2A:81 SRSF1:163 NARS1:230 DTYMK:349 RARS1:353 CARM1:452
REACTOME_SPERM_MOTILITY_AND_TAXES	0.42619112	8	2.986e-05	9.685e-03	CATSPERB:106 CATSPER3:403 KCNU1:958 CATSPERD:1007 CATSPERG:1423 CATSPER4:1441
REACTOME_GLUCCOSE_METABOLISM	0.13321519	81	3.448e-05	1.065e-02	HK2:42 SEHL1:309 GNPDAA2:494 GOT1:510 HK1:523 GOT2:546
DARWICHE_SQUAMOUS_CELL_CARCINOMA_UP	0.11060910	117	3.661e-05	1.079e-02	DYNLL1:6 ENG:22 EEF1B2:121 SRSF1:163 RPL23:7 HK2:42 RACK1:109
SEIDEN_ONCOGENESIS_BY_MET	0.14004251	72	4.022e-05	1.134e-02	TSPAN8:28 ADAM9:65 HAT1:260 UGDH:565 HIGD1A:840 EPS8:1044
MARTORIATI_MDM4_TARGETS_FETAL_LIVER_DN	0.05758122	432	4.417e-05	1.146e-02	FGG:30 TNPO3:47 BNC1:70 COL1A1:123 MAMC12:139 SRSF1:163
WP_MRINA_ROLE_IN_IMMUNE_RESPONSE_IN_SEPS	0.21216185	31	4.359e-05	1.146e-02	TNF:189 TLR7:279 TLR8:334 VCAM1:370 ICAM1:431 RBL1:603
BURTON_ADIPOGENESIS_6	0.09094916	163	6.336e-05	1.522e-02	HSD11B1:225 CA3:330 S100A4:448 COX5B:535 ORM2:626 MEI1:641
KEGG_GLUTATHIONE_METABOLISM	0.19028553	37	6.218e-05	1.522e-02	GGT5:320 OPLAH:329 IDH1:612 GGT6:744 MGST3:823 GSTO1:850
DIAZ_CHRONIC_MYELOGENOUS_LEUKEMIA_UP	0.03515991	1149	7.362e-05	1.706e-02	R3DHMA:2 SULT1A2:9 DNMT3B:18 UBE2B:23 HIRA:29 SNRPD1:31
REACTOME_ACTIVATION_OF_THE_MRNA_UPON_BIN	0.21467407	28	8.453e-05	1.891e-02	RPS14:17 RPS15A:79 RPS27:207 RPS7:374 RPS26:481 RPS15:597
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	0.04878556	555	9.402e-05	1.967e-02	CD80:27 UBE2D1:37 SYK:48 CD86:55 TNFSF8:111 CASP11:112
REACTOME_METABOLISM_OF_LIPIDS	0.04528549	650	9.222e-05	1.967e-02	MVK:33 HSD17B2:62 PI4K2A:73 CYP51A1:75 UBE2I:94 RAB14:113
REACTOME_GLUTATHIONE_CONJUGATION	0.22349256	25	1.099e-04	2.228e-02	GGT5:320 OPLAH:329 GSTT2B:723 GGT6:744 CHAC1:750 MGST3:823
REACTOME_SARS_COV_2_MODULATES_HOST_TRANS	0.21835992	26	1.165e-04	2.289e-02	RPS14:17 SNRPD1:31 RPS15A:79 RPS27:207 RPS7:374 RPS26:481
DARWICHE_PAPILLOMA_RISK_LOW_UP	0.09949041	124	1.326e-04	2.458e-02	DYNLL1:6 ENG:22 EEF1B2:121 SRSF1:163 RPL23:7 COL20A1:325
TIEN_INTESTINE_PROBIOTICS_24HR_UP	0.05232673	462	1.292e-04	2.458e-02	DYNLL1:6 DNMT3B:18 SNRPD1:31 UBE2D1:37 TMEM50A:52 ETF1:59
REACTOME_SARS_COV_INFECTIONS	0.06238650	312	1.590e-04	2.761e-02	ANOS1:4 RPS14:17 SNRPD1:31 SYK:48 RPS15A:79 GATAD2A:81
LEE_LIVER_CANCER_SURVIVAL_DN	0.09581378	131	1.557e-04	2.761e-02	DYNLL1:6 UBE2D1:37 CALM2:222 OPLAH:329 H4C14:30 RPS7:374
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.20600036	28	1.617e-04	2.761e-02	HSD17B2:62 CYP17A1:145 HSD11B1:225 CYP19A1:506 COMT:547 HSD17B3:600
REACTOME_DRUG_ADME	0.13676973	63	1.752e-04	2.914e-02	SULT1A3:21 SULT1A4:24 HSD11B1:225 GGT5:320 SLC29A2:665 SLC01A2:678
KEGG_PHENYLALANINE_METABOLISM	0.26236093	17	1.804e-04	2.926e-02	PRDX6:479 GOT1:510 GOT2:546 PAH:889 TAT:1042 MIF:1333

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MYELOPROLIFERATIVE_DISORDER, CHRONIC, WI	0.21711512	38	3.678e-06	1.657e-02	TNF:189 CXADR:223 IFNG:226 CXCR2:251 ICAM1:431 COMT:547
Hyperglycemia	0.06684943	416	3.472e-06	1.657e-02	TRPV1:19 HK2:42 JUND:88 FGF21:104 CASP3:112 TNFSF10:127
Lymphoid leukemia	0.10056624	174	5.071e-06	1.657e-02	RPS14:17 SYK:48 PRMT5:150 FLT3:157 TNF:189 IFNG:226
Malignant tumor of colon	0.03518899	1540	7.922e-06	1.693e-02	PROM1:4 TRA2B:8 DNMT3B:18 CD80:27 HK2:42 CEMIP:45
Pneumonia	0.06586816	393	8.635e-06	1.693e-02	DNMT3B:18 FGG:30 CEMIP:45 SYK:48 CD86:55 ICOS:76
Inflammatory dermatosis	0.11197252	130	1.090e-05	1.782e-02	FLT3:157 TNF:189 IFNG:226 CXCR2:251 GRN:265 TLR7:279
Ichthyosis, X-Linked	0.13944686	80	1.662e-05	1.811e-02	CASP3:112 TNFSF10:127 AITF:174 TNF:189 DMD:244 MBI1:387
Osteoporosis, Senile	0.16593220	57	1.496e-05	1.811e-02	PGLS:87 COL1A1:123 CALCR:144 HSD11B1:225 U2AF1:231 CYP19A1:506
Dengue Fever	0.11344480	123	1.452e-05	1.811e-02	SYK:48 TNFSF10:127 LTA:182 TNF:189 HSPA5:197 IFNG:226
Thyroid associated ophthalmopathies	0.14479881	73	1.933e-05	1.895e-02	CD80:27 CD86:55 LTA:182 TNF:189 ST8SIA1:196 IFNG:226
Thromboangiitis Obliterans	0.19405813	40	2.192e-05	1.954e-02	TNF:189 ST8SIA1:196 IFNG:226 VCAM1:370 ICAM1:431 RBM45:512
Gingival Diseases	0.12745011	92	2.461e-05	2.011e-02	CD80:27 CLU:116 SNAP25:143 TNF:189 IFNG:226 IL4:322
Autoimmune Diseases	0.04411395	790	3.299e-05	2.423e-02	RPL23:7 CD80:27 TNPO3:47 AFP:49 CD86:55 GGTLC3:68
Small head	-0.05419296	508	3.460e-05	2.423e-02	BCOR:3 MATN4:23 PCNT:33 RBL1:64 ALDH18A1:97 ATRX:102
Malaria, Cerebral	0.14075266	69	5.382e-05	2.989e-02	LTA:182 TNF:189 IFNG:226 TLR7:279 IL33:297 IL4:322
Myeloid Leukemia, Chronic	0.04725737	636	5.873e-05	2.989e-02	PROM1:4 DNMT3B:18 CD80:27 SCPEP1:46 SYK:48 CD86:55
Adult Hepatocellular Carcinoma	0.12713940	83	6.376e-05	2.989e-02	DNMT3B:18 ENG:22 SAV1:105 SRSF1:163 AKR1B1:271 GJB2:427
Chronic inflammatory disorder	0.19272626	36	6.345e-05	2.989e-02	TNF:189 CXCR2:251 CARD14:275 IL33:297 TLR9:443 IL1RN:537
Colitis	0.05300791	499	5.923e-05	2.989e-02	TRPV1:19 CD86:55 ICOS:76 ROS1:83 COL2A1:108 TNFSF8:111
Nonalcoholic Steatohepatitis	0.08966150	168	6.403e-05	2.989e-02	DNMT3B:18 FGF21:104 CASP3:112 COL1A1:123 TNFSF10:127 CYP17A1:145
Vasculitis	0.12114650	93	5.548e-05	2.989e-02	MVK:33 CXCL13:126 TNF:189 IFNG:226 AKR1B1:271 VCAM1:370
Microcephaly	-0.08372857	191	6.996e-05	3.118e-02	PCNT:33 ATRX:102 UBE3B:113 MED17:132 CENPJ:139 MYO16:166
Anemia	0.05871318	390	7.786e-05	3.319e-02	RPS14:17 DNMT3B:18 ENG:22 ATP2B4:54 CASP3:112 TNFSF10:127